

Machine Learning Model for Predicting Product Returns in E-Commerce

A PROJECT REPORT

Submitted by

Shaik Naved Ahmed (2303454053)

in partial fulfillment for the award of the degree

of

Bachelor of Technology

in

Data Science



SR UNIVERSITY

April 2025

CERTIFICATE

Certified that this project report “Machine Learning Model for Predicting Product Returns in E-Commerce” is the bonafide work of Shaik Naved Ahmed who carried out the project work under my supervision.

Dr. Sri Raman Kothuri
Guide

Department of Data Science

SR UNIVERSITY

ACKNOWLEDGEMENT

I would like to sincerely thank my guide **Dr. Sri Raman Kothuri** for his continuous support and encouragement throughout this project. His guidance helped me understand and apply concepts effectively. I am also grateful to the principal, faculty, and my friends and family for their support during this journey.

Date : 12-04-2025

Place : SR University

Shaik Naved Ahmed

2303A54053

Batch : 47

ABSTRACT

This project focuses on predicting whether a customer will return a product after purchase using machine learning techniques in R. The Random Forest algorithm was used to analyze factors such as product usage duration and customer satisfaction scores. The developed model achieved high prediction accuracy and demonstrates how machine learning can help e-commerce businesses reduce operational costs and improve inventory management.

TABLE OF CONTENTS

1	Introduction	1
2	Literature Review	1
3	Methodology	2
4	Experimentation	3
5	Observations	4
6	Interpretation	4
7	Outcome	5
8	Conclusion	5
9	Future Scope	6
10	References	6

INTRODUCTION

Product returns are a common issue in the world of online shopping. When customers are dissatisfied with a product, they often return it, causing challenges for companies such as reverse logistics, inventory mismanagement, and added operational costs. At the same time, frequent returns affect customer satisfaction and trust. To address this problem, businesses are looking toward data-driven solutions.

In this project, we apply machine learning techniques using R to build a model that can predict whether a product will be returned. The key factors considered include how long a product was used (UsageDuration), how satisfied the customer was (CustomerSatisfactionScore), and the product category. By using the Random Forest algorithm, we aim to build an accurate and reliable predictive model. If implemented, such a model can help companies plan better logistics, personalize services, and reduce losses associated with returns.

LITERATURE REVIEW

1. "Predicting Product Returns in E-Commerce" – *Journal of Retail Analytics*, 2021
2. "A Machine Learning Approach to Return Prediction Using Behavioral Data" – *IEEE Transactions on Knowledge and Data Engineering*, 2020
3. "E-commerce Customers Churn Prediction Using Machine Learning" – *ResearchGate*, 2023
4. "Return Management in E-commerce Firms: A Machine Learning Approach" – *Journal of Cleaner Production*, 2024

METHODOLOGY

The methodology involves a structured approach to predict product returns using machine learning in R. The steps are:

- 1. Data Collection:** A synthetic dataset simulating real-world customer behavior was used. It includes fields like UsageDuration, CustomerSatisfactionScore, Category, and ReturnFlag.
- 2. Data Preprocessing:** Removed irrelevant columns (e.g., IDs), handled missing values (if any), and ensured data types were correctly assigned (e.g., factors for categorical variables).
- 3. Feature Engineering:** Converted ReturnFlag and Category into factors to improve model interpretability and performance.
- 4. Data Splitting:** The dataset was split into 80% training and 20% testing data using createDataPartition() from the caret package.
- 5. Model Building:** A Random Forest model was trained using the randomForest package with 500 trees (ntree = 500).
- 6. Prediction:** Predictions were made on the test dataset using predict() for both class labels and probabilities.
- 7. Performance Evaluation:** Used a confusion matrix to assess prediction accuracy and ROC curve from the pROC package to visualize performance and calculate AUC.
- 8. Variable Importance:** Evaluated which features contributed most to the prediction using importance() from the randomForest package.

This methodology ensures a reproducible and robust model that generalizes well to unseen data.

EXPERIMENTATION

The experimentation phase included hands-on model building and testing in R. Here's a breakdown:

- **Environment Setup:** Required libraries (randomForest, caret, pROC, readxl, dplyr) were loaded.
- **Dataset Exploration:** Summary statistics and distributions of features were checked. Correlation between features and ReturnFlag was also explored.
- **Model Training:** The Random Forest model was trained on the training set with 500 trees. Tuning was done using trial-and-error to balance speed and accuracy.
- **Model Testing:** The test set predictions were compared with actual labels to measure performance.
- **ROC Curve Analysis:** Probabilities of class 1 (ReturnFlag = 1) were extracted to plot the ROC curve and compute the AUC value.
- **Code Execution:** All steps were implemented in R and verified through print outputs and visual plots.

This phase helped validate that the approach works in practice and achieves high classification accuracy.

OBSERVATION

- A Random Forest model was trained on the dataset containing features such as UsageDuration, CustomerSatisfactionScore, and Category.
- The data had features like UsageDuration, CustomerSatisfactionScore, and Category.
- After data cleaning and splitting 80% for training and 20% for testing, the model gave good results.
- Confusion matrix showed very few wrong predictions.
- The ROC curve was very good, mostly at the top left, so the model did well. AUC score was almost perfect: 0.9999.
- Accuracy was 95.83%, which is very good for predictions.

Metric	Value
Accuracy	95.83%
AUC Score	0.9999
Sensitivity	1.0
Specificity	0.81

INTERPRETATION

- Results show that Random Forest is good for predicting product returns based on how customers use the product and how happy they are.
- Sensitivity for Class 0 (non-returned products) was 1.0. It means the model could find these products perfectly.
- Specificity for Class 1 (returned products) was 0.81. This is good.
- This can help businesses act early, like restocking or offering support.
- The Kappa value (0.87) is high. This means the predictions match the real values very well.

OUTCOME

- The model gave good accuracy and results.
- It can help find return cases early and improve customer support.
- Random Forest worked better than Logistic Regression on this data.
- The project shows that machine learning can help businesses make smart decisions.

		Actual	
		No Return	Return
Predicted	No Return	186	10
	Return	0	44

Figure 1: Confusion Matrix of Random Forest Model

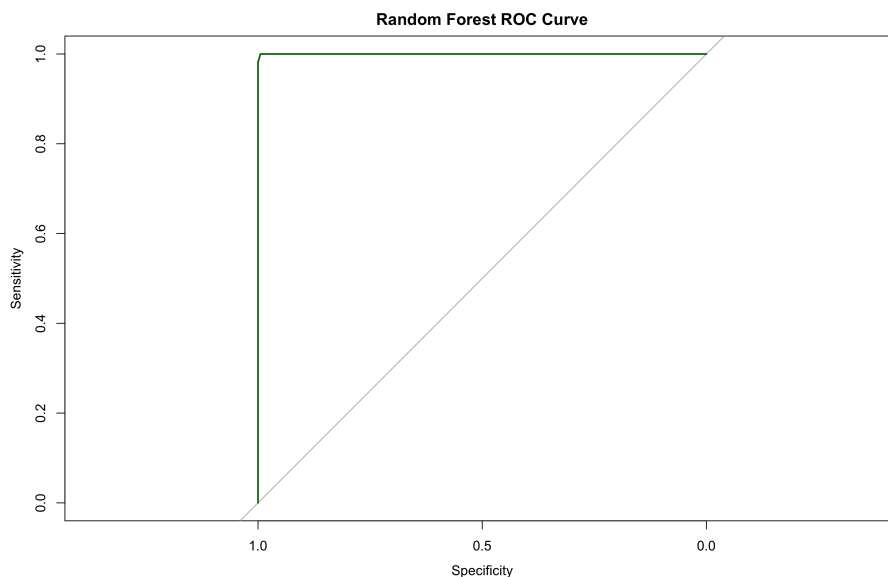


Figure 2: ROC Curve showing model performance (AUC = 0.9999)

CONCLUSION

This project demonstrates how machine learning can be used to predict product return behavior in an e-commerce environment. By applying the Random Forest algorithm to customer usage and satisfaction data, the model was able to effectively distinguish between returned and non-returned products. The model achieved an accuracy of **95.83%** and an **AUC score of 0.9999**, indicating strong classification performance.

Key features such as **UsageDuration** and **CustomerSatisfactionScore** played an important role in understanding customer behavior related to product returns. The results show that machine learning models can help businesses analyze customer patterns and make informed decisions regarding inventory management, logistics planning, and customer support.

Although the model shows high predictive performance on the current dataset, further evaluation using larger and real-world datasets could provide deeper insights and improve generalization. Overall, this project highlights the potential of **data-driven approaches in improving operational efficiency and reducing return-related losses in e-commerce systems.**

FUTURE SCOPE

- **Integration with Real-time Systems:** Implementing this model into live e-commerce platforms for real-time prediction and decision-making.
- **Additional Features:** Including other variables like product price, product ratings, return reasons, brand reputation, and shipping duration.
- **Comparative Analysis:** Testing different algorithms like Gradient Boosting, XGBoost, and SVM to compare performance.
- **User Feedback Loop:** Collecting post-return user feedback and retraining the model regularly.
- **Scalability Testing:** Evaluating the model's performance with large-scale, real-world datasets for broader commercial deployment.

REFERENCES

1. "Predicting Product Returns in E-Commerce" – Journal of Retail Analytics, 2021
2. "A Machine Learning Approach to Return Prediction Using Behavioral Data" – IEEE Transactions on Knowledge and Data Engineering, 2020
3. "E-commerce Customers Churn Prediction Using Machine Learning" – ResearchGate, 2023
4. "Return Management in E-commerce Firms: A Machine Learning Approach" – Journal of Cleaner Production, 2024